

Networking Services

Prof. Arnab Mandal

Assistant Professor
CSE Department

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IP Addressing

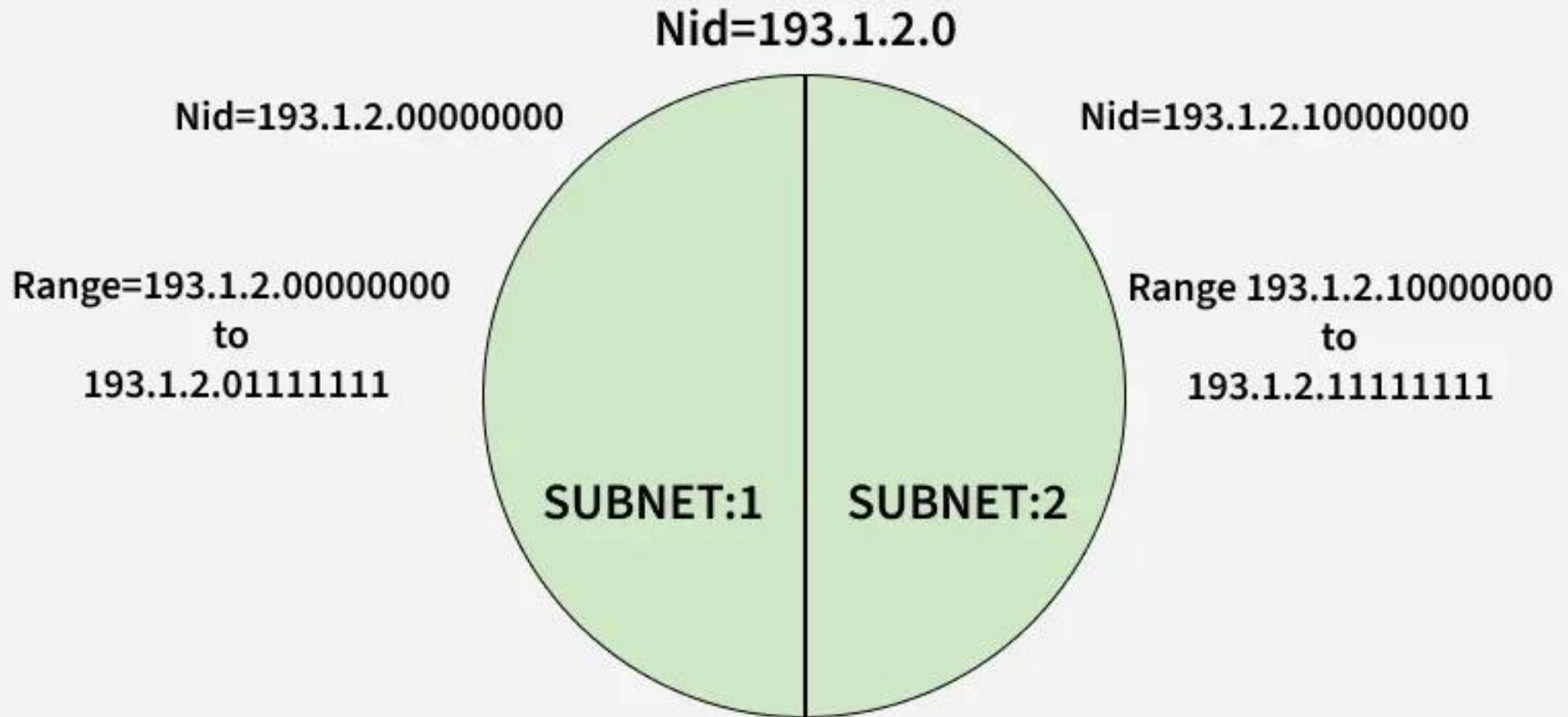
- IP Address An Internet Protocol (IP) address is a unique numerical label assigned to each device connected to a computer network.
- IPv4: The most common format, consisting of four numbers (0 255) separated by dots (e.g., 192.168.1.1). It uses 32-bit addressing, providing ~4.3 billion addresses.
- IPv6: Created to replace IPv4 due to address exhaustion. It uses 128-bit hexadecimal (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- Public IP: Reachable from the entire internet.
- Private IP: Used within a local network (LAN) and not routable on the public internet.

CIDR (Classless Inter-Domain Routing)

- CIDR is a method for allocating IP addresses and IP routing.
- It replaced the old "Class A, B, C" system to be more efficient.
- Notation: It looks like an IP address followed by a slash and a number (the prefix), such as 10.0.0.0/16.
- The Slash (mask): The number after the / represents how many bits are "fixed" for the network portion.
 - /32: Exactly one IP address.
 - /24: 256 IP addresses (e.g., 192.168.1.0 to 192.168.1.255).
 - /16: 65,536 IP addresses.
- Rule of Thumb: The smaller the CIDR number, the larger the number of available IP addresses.

Subnets

- A Subnet (sub-network) is a logical subdivision of an IP network. Dividing a large network into smaller subnetworks improves performance and security.
- Public Subnet: Has a route to an Internet Gateway (accessible from the internet).
- Private Subnet: No direct route to the internet; usually houses databases or internal backend logic.
- Efficiency: Subnetting prevents "broadcast storms" where too many devices try to communicate at once, slowing down the entire network.



Feature	Subnet 1	Subnet 2
Borrowed Bit	0XXXXXXXX	1XXXXXXXX
New Network ID	193.1.2.0	193.1.2.128
Binary Range	00000000 to 01111111	10000000 to 11111111

The First Address: Network ID

The first IP in the range (where all host bits are 0) is the Network Identifier.

Purpose: It represents the network itself.

The Last Address: Broadcast Address

The last IP in the range (where all host bits are 1) is the Broadcast Address.

Purpose: It is used to send data to every device on that subnet simultaneously.

How it works: If a device sends a packet to this address, the network switch ensures every other device in that subnet receives it. This is often used for discovery (like a computer looking for a printer).

Example: In the subnet 192.168.1.0/24, the address 192.168.1.255 is the broadcast address.

The "Usable" IP Formula

Because of these two reserved addresses, the number of devices you can actually connect to a subnet is always minus 2.

The formula for usable hosts is:

$$2^n - 2$$

(Where n is the number of host bits)

Example: A /24 Subnet

Total addresses: $2^8 = 256$

Reserved: 2 (Network ID and Broadcast)

Usable IPs: $256 - 2 = 254$

Note for Cloud Users: If you are working in AWS VPCs, they actually reserve five addresses per subnet (the first four and the last one) for things like the VPC router, DNS, and future use!

Subnet Mask :

A [subnet mask](#) is a 32-bit number used in IP addressing to separate the network portion of an [IP address](#) from the host portion. It helps computers and devices determine which part of an IP address refers to the network they are present, and which part refers to their specific location or address within that network.

Subnet Mask

11111111 . 11111111 . 11111111 . 00000000

255 . 255 . 255 . 0

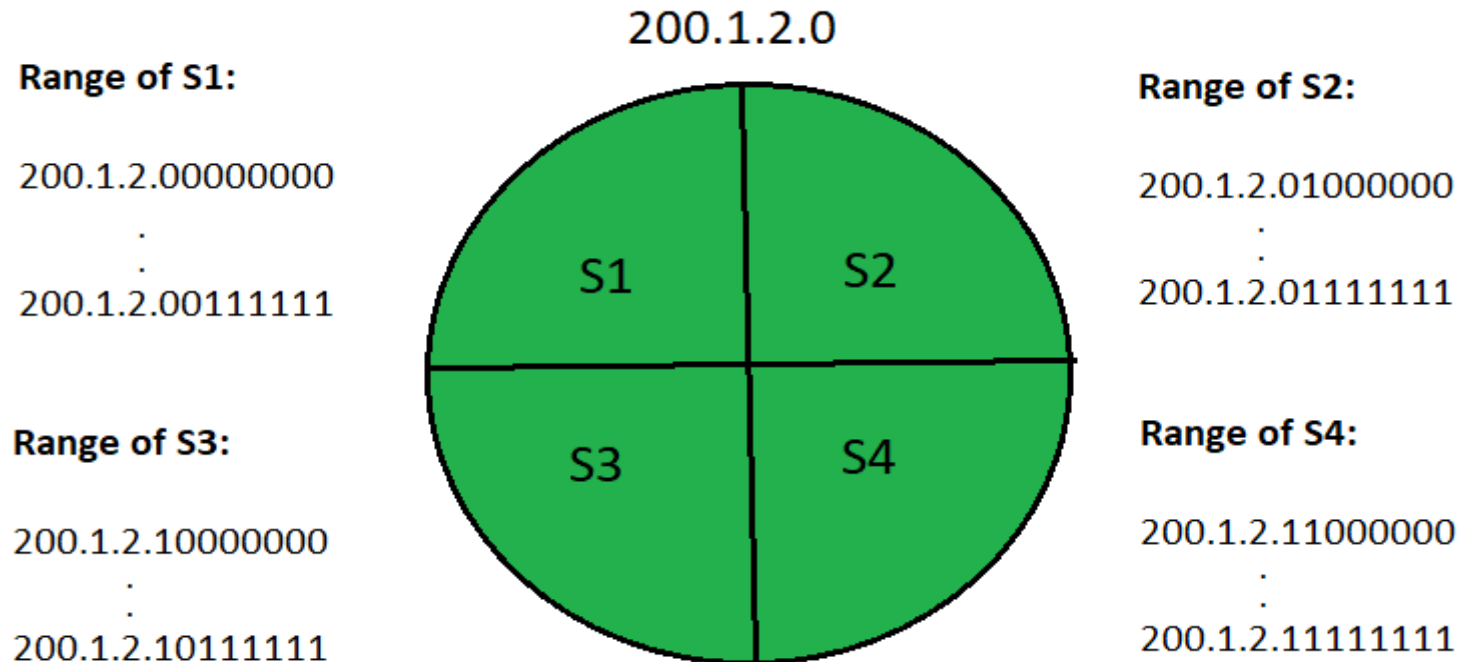
255.255.255 . 0

Network

Host

Routing Tables :

A routing table is a set of rules, often viewed in table format, that is used to determine where data packets traveling over an Internet Protocol (IP) network will be directed. All IP-enabled devices, including routers and switches, use routing tables.





Destination	Subnet Mask	Interface
200.1.2.0	255.255.255.192	a
200.1.2.64	255.255.255.192	b
200.1.2.128	255.255.255.192	c
200.1.2.192	255.255.255.192	d
Default	0.0.0.0	e



Let's assume a packet arrives with the destination IP: 200.1.2.75.

We need to check which interface (a, b, c, d, or e) it should use.

Step 1: Convert to Binary

To perform the ANDing, we convert the relevant parts of the IP and the Mask to binary. Since the first three octets (200.1.2) match in our table, we only need to focus on the last octet.

- Destination IP (.75): 01001011
- Subnet Mask (.192): 11000000 (This is a /26 prefix) (Why .192? because I have selected two bits from the host id part to break the entire network into 4 subnets. So, I place two 1's in the last octet for creating the Subnet mask)

Step 2: The ANDing Operation

The logical AND rule is simple: 1 AND 1 = 1; everything else is 0.



Operation	Binary Calculation	Result (Decimal)
IP (.75)	0 1 0 0 1 0 1 1	
Mask (.192)	1 1 0 0 0 0 0 0	
Result (AND)	0 1 0 0 0 0 0 0	.64



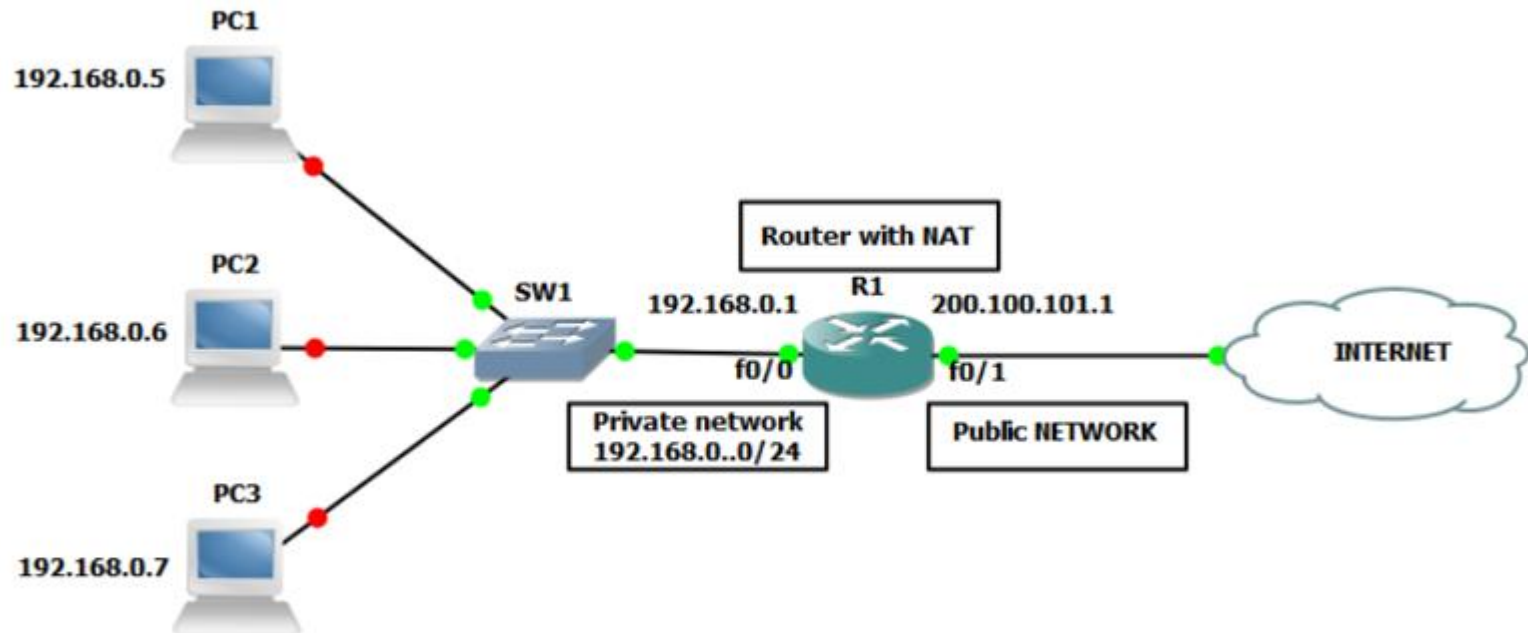
The result of the computation is 200.1.2.64.

Step 3: Match the Table

Now the router compares that result against the "Destination" column in your table:

1. Check Row 1: Does .64 match 200.1.2.0? No.
2. Check Row 2: Does .64 match 200.1.2.64? YES.
3. Action: The router stops here and forwards the packet to Interface b.

NAT(Network Address Translation)



The Internal (Private) Network

On the left, we have three computers (PC1, PC2, PC3) sitting behind a switch (SW1).

Private IP Range: They are part of the 192.168.0.0/24 subnet.

Individual IPs: Each PC has a unique private IP (e.g., 192.168.0.5).

The Gateway: The router's internal interface (f0/0) has the IP 192.168.0.1. This acts as the "door" for the PCs to leave their local network.

2. The Router (R1) as the Translator

The router is the "Middleman" between the private and public worlds. It has two "faces":

Inside Interface (f0/0): Connected to the private network.

Outside Interface (f0/1): Connected to the Public Network (the Internet) with the IP 200.100.101.1.

3. The NAT Process in Action

If PC1 (192.168.0.5) wants to visit a website on the Internet:

Request: PC1 sends a packet destined for the Internet.

Intercept: The router (R1) receives this packet. It knows that the private IP 192.168.0.5 cannot travel on the public internet (it would be dropped).

Translation: The router replaces PC1's private IP with its own Public IP (200.100.101.1) and assigns a unique port number to the request.

Tracking: The router records this in a NAT Table (e.g., "PC1 is currently using Port 5000").

Response: When the Internet sends data back, it sends it to 200.100.101.1. The router looks at the port number, checks its table, and realizes, "Ah, this belongs to PC1!" and passes the data back to 192.168.0.5.

VPC(Virtual Private Network)

A VPC (Virtual Private Cloud) is a private, isolated section of a public cloud (like AWS, Google Cloud, or Azure) where you can launch resources in a virtual network that you define.

Think of a Public Cloud as a massive, shared hotel. A VPC is like renting an entire floor: you have your own hallway, you decide who gets a key, and you control which rooms (servers) can talk to each other.

Core Components of a VPC

To build a functional VPC, you need the following components working together:

IPv4/IPv6 CIDR Block: The range of IP addresses for your VPC (e.g., 10.0.0.0/16).

Subnets: Slicing your VPC into smaller address ranges (e.g., Public Subnet for web servers, Private Subnet for databases).

Route Tables: A set of rules (the "GPS") that tells network traffic where to go.

Internet Gateway (IGW): The "front door" that connects your VPC to the open internet.

NAT Gateway: Allows resources in a private subnet to "call out" to the internet (for updates) without letting the internet "call in."

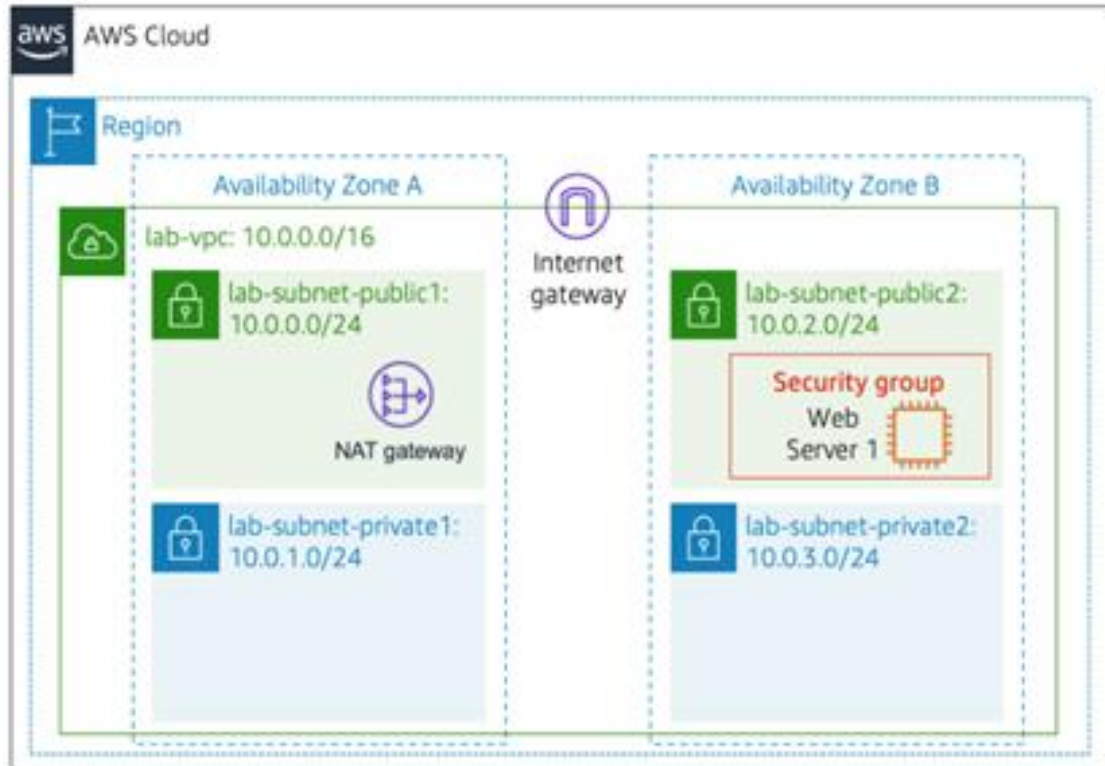
Why Use a VPC?

Isolation: Your data is logically separated from every other customer in the cloud.

Customization: You select your own IP range, create your own subnets, and configure your own route tables.

Security: You can hide sensitive databases in private subnets with no path to the internet.

Hybrid Connectivity: You can connect your VPC to your physical on-premise office using a VPN or a dedicated fiber connection (Direct Connect).



Public Route Table

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	Internet gateway

Private Route Tables

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	NAT gateway

The logically isolated network in the AWS cloud.

Defined by a CIDR block (e.g., 10.0.0.0/16).

It functions like your own private data center in the cloud.

2. Subnets (The Zones): As a Dev ops engineer I break the entire VPC network by setting the prefix as 10.0.0.0/24 that means I have taken 8 bits from the host Id part So total 256 subnets are possible .

Segments of the VPC's IP range.

Public Subnet: Connected to the Internet (Web Servers).

Private Subnet: Hidden from the Internet (Databases).

High Availability: We put subnets in different Availability Zones (AZs) so if one AZ's loses power, the other stays up.

3. Route Tables

A Route Table is the digital "GPS" for your subnet. It contains a list of rules that tell data packets exactly which path to choose to reach their destination.

Destination (Where): The IP address the packet is trying to visit (e.g., 10.0.0.5 or google.com).

Target (How): The specific gateway or connection used to get there (e.g., Local, Internet Gateway, or NAT).

Destination	Target	In Plain English
10.0.0.0/16	Local	"If the packet is for another building on campus, stay inside."
0.0.0.0/0	IGW	"If the packet is for the Internet, go to the Main Gate."

4. Internet Gateway (IGW)

The "bidirectional" door. It allows the VPC to talk to the internet and the internet to talk back. Essential for Public Subnets.

5. NAT Gateway

Allows instances in Private Subnets to reach the internet (for updates) but prevents the internet from starting a connection with them. NAT always sits in a Public Subnet.

6. Load Balancer

The single point of entry that distributes incoming traffic across multiple servers to prevent crashes and ensure high speed.

There are various types of Load Balancer:

a. Application Load Balancer (ALB): The ALB is the most common. It operates at the Application Layer and can read the content of the request (HTTP/HTTPS). How it works: It can look at the URL. For example, if a student goes to `parul.ac.in/fees`, the ALB sends them to the Finance Building. If they go to `parul.ac.in/library`, it sends them to the Library Building.
Best For: Web applications and microservices where you need "Path-based" or "Host-based" routing.

b. Network Load Balancer (NLB): The NLB operates at the Transport Layer (TCP/UDP). It doesn't "read" the data; it just sees the IP address and Port and shunts the traffic at lightning speed.

How it works: It is built for extreme performance. It can handle millions of requests per second with ultra-low latency.

Best For: Gaming apps, video streaming, or any service where raw speed is more important than "smart" routing. It also provides a Static IP, which the ALB does not.

c. Gateway Load Balancer (GWLB): This is a specialized load balancer used for "Inspecting" traffic before it even enters your main campus area.

How it works: It sits in front of your VPC and sends all incoming traffic to a "Fleet" of third-party security appliances (like Firewalls or Intrusion Detection Systems). Once the traffic is cleared, it passes it into the VPC.

Best For: Deep security inspection and managing a "wall" of firewalls easily.

7. Security Groups & NACLs

Security Group: Firewall for the Instance (Stateful—remembers you).

NACL: Firewall for the Subnet (Stateless—checks everyone every time).

Packet Flow Example

If a user tries to access your website:

Entry: The request hits the Internet Gateway.

Routing: The Public Route Table directs the request to the Public Subnet in AZ B.

Final Security Check: The Security Group checks if the request is allowed. If yes, it reaches Web Server 1.

Backend Access: Web Server 1 communicates locally with a database in the Private Subnet to fetch data.

Return: The data flows back to the user through the same path.

VPC Security:

A **Virtual Private Cloud (VPC)** is your isolated section of the AWS Cloud. Security within a VPC is handled at two primary layers:

Security Groups (Instance Level)

- **What they are:** Virtual firewalls for your **EC2 instances**.
- **Behavior: Stateful.** If you allow an incoming request (e.g., on port 80), the outgoing response is automatically allowed, regardless of outbound rules.
- **Default:** Denies all inbound traffic and allows all outbound traffic.

Network ACLs (Subnet Level)

What they are: An optional layer of security for your **subnets**.

Behavior: Stateless. You must explicitly define both inbound and outbound rules for traffic to flow.

Use Case: Broad "block" rules (e.g., blocking a specific malicious IP range across an entire subnet).

DNS: AWS Route 53

Route 53 is a highly available and scalable Domain Name System (DNS) web service. It "routes" users to internet applications by translating human-readable names (like `www.example.com`) into numeric IP addresses.

Domain Registration: You can buy and manage domain names directly through AWS.

Health Checks: Route 53 can monitor your application's health. If a server goes down, it automatically redirects traffic to a healthy one.

Routing Policies:

Simple: One domain points to one IP.

Latency-based: Routes users to the AWS Region that provides the lowest latency.

Failover: Used for disaster recovery to route traffic to a secondary "backup" site.

Amazon CloudFront (CDN)

CloudFront is a Content Delivery Network (CDN) that speeds up the distribution of your static and dynamic web content (like .html, .css, .js, and images).

How it works: It uses a global network of Edge Locations. When a user requests content, CloudFront delivers it from the edge location with the lowest latency.

Origin Shield: Reduces the load on your web server (the "Origin") by adding a centralized caching layer.

Security: Integrated with AWS WAF (Web Application Firewall) to protect against common web exploits like SQL injection.

Launch a Web Server in a VPC

Step A: Prepare the VPC

- **Create a VPC:** Define your private IP address range (e.g., 10.0.0.0/16).
- **Create a Subnet:** Create a public subnet (e.g., 10.0.1.0/24) where your web server will live.
- **Internet Gateway (IGW):** Attach an IGW to your VPC to allow communication with the internet.
- **Route Table:** Update the subnet's route table to point internet-bound traffic (0.0.0.0/0) to the IGW.

Step B: Launch the EC2 Instance

Choose an AMI: Select "Amazon Linux 2023" (Free Tier eligible).

Instance Type: Choose t3.micro or t2.micro.

Network Settings: Select your VPC and Public Subnet. Ensure Auto-assign Public IP is enabled.

Security Group: Create a rule to allow HTTP (Port 80) from 0.0.0.0/0.

Step C: Configure the Web Server

Once the instance is running, connect via EC2 Instance Connect and run:

```
sudo yum update -y  
sudo yum install -y httpd  
sudo systemctl start httpd  
sudo systemctl enable httpd  
echo "<h1>Hello from my AWS Web Server!</h1>" > /var/www/html/index.index.html
```

Key Takeaways:

- **VPC (Virtual Private Cloud):** Acts as your own isolated, private section of the AWS Cloud where you define your own IP address range and network configuration.
- **Subnets:** Segments of your VPC divided into Public Subnets (accessible to the internet) and Private Subnets (hidden from the internet for databases or internal services).Route
- **Tables:** A set of rules (routes) that determine where network traffic from your subnet or gateway is directed.
- **Security Groups:** Act as a virtual stateful firewall for your EC2 instances, controlling both inbound and outbound traffic at the instance level.VPC Security: Enhanced by Network ACLs (NACLs), which act as a stateless firewall at the subnet level, providing a second layer of defense.

- **Route 53:** A highly available and scalable DNS (Domain Name System) web service that connects user requests (like `www.example.com`) to AWS resources like EC2 or S3.
- **CloudFront:** A Content Delivery Network (CDN) that speeds up the delivery of static and dynamic web content by caching it at "Edge Locations" closer to your users.
- **VPC Lab:** To launch a web server, you must create a VPC, attach an Internet Gateway to the public subnet, and configure a Security Group to allow HTTP (port 80) traffic.

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